

Quantum Systems in Cavities: Spectral properties of an SSH chain coupled to a cavity mode

Valentina C. Barroso^{1,2}, Pablo S. Cornaglia^{1,2}, Gonzalo Usaj^{1,2}

¹ Centro Atómico Bariloche and Instituto Balseiro (Universidad Nacional de Cuyo), 8400 San Carlos de Bariloche, Río Negro, Argentina.

² Instituto de Nanociencia y Nanotecnología (INN), CONICET-CNEA, Argentina.



Abstract

In this work, we investigate the coupling between a cavity mode of energy $\hbar\Omega$ and fermions in a Su-Schrieffer-Heeger (SSH) chain. Our results show that the π -gap present in the semi-classical Floquet limit persists for all values Ω that lead to a resonant process only for the single-fermion case; for the many fermions case, the gap closes when $\hbar\Omega > 2E_F$ due to Pauli blocking. Finally, we characterize the formation of anomalous edge states in this system, explicitly showing that the photonic spectral density reveals their presence.

Theoretical Model

SSH Chain in a cavity

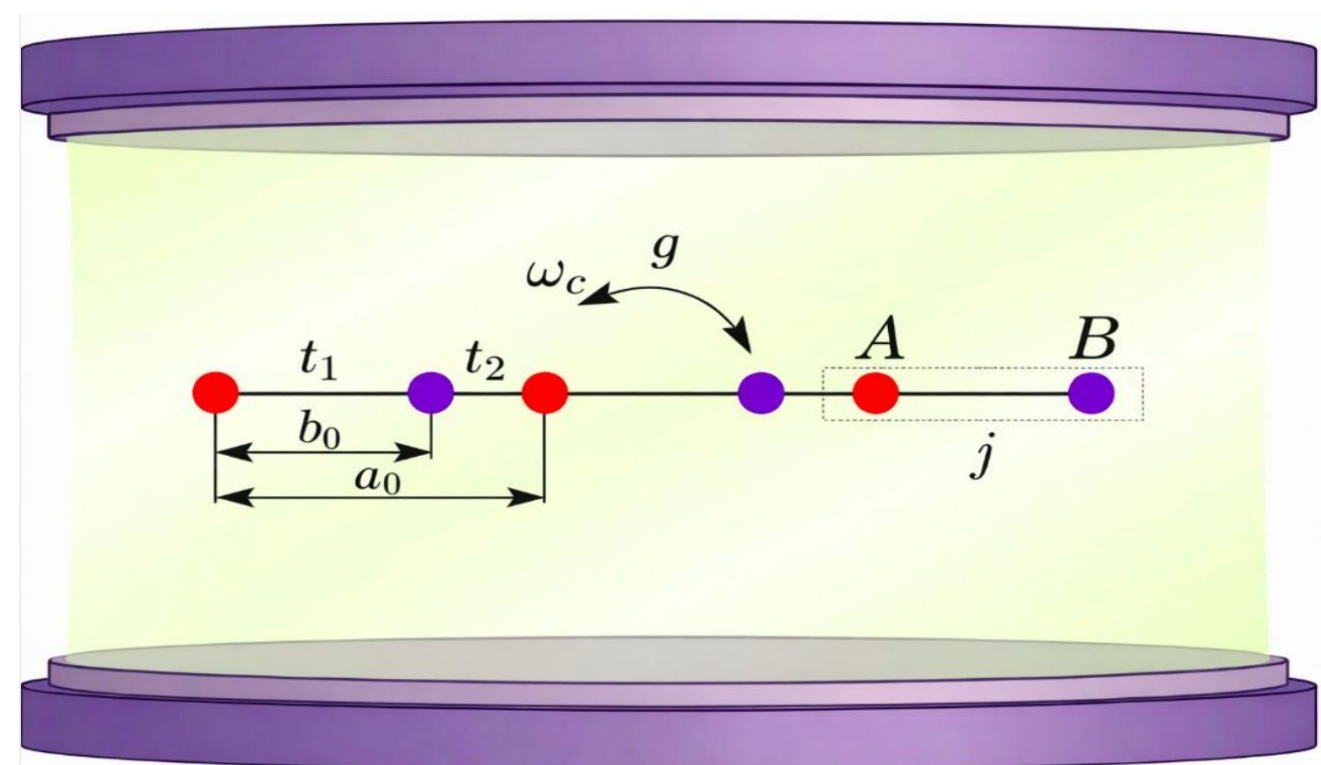
The hybrid SSH-cavity system is governed by a tight-binding Hamiltonian with single-mode cavity coupling:

$$H = \tilde{t}_1 \sum_{j=1}^L (c_{j,A}^\dagger c_{j,B} + \text{h.c.}) + \tilde{t}_2 \sum_{j=1}^{L-1} (c_{j+1,A}^\dagger c_{j,B} + \text{h.c.}) + \omega_c \left(a^\dagger a + \frac{1}{2} \right)$$

The hopping amplitudes $\tilde{t}_1$ and $\tilde{t}_2$ are modulated by the boson field.

Peierls Phase Substitution :

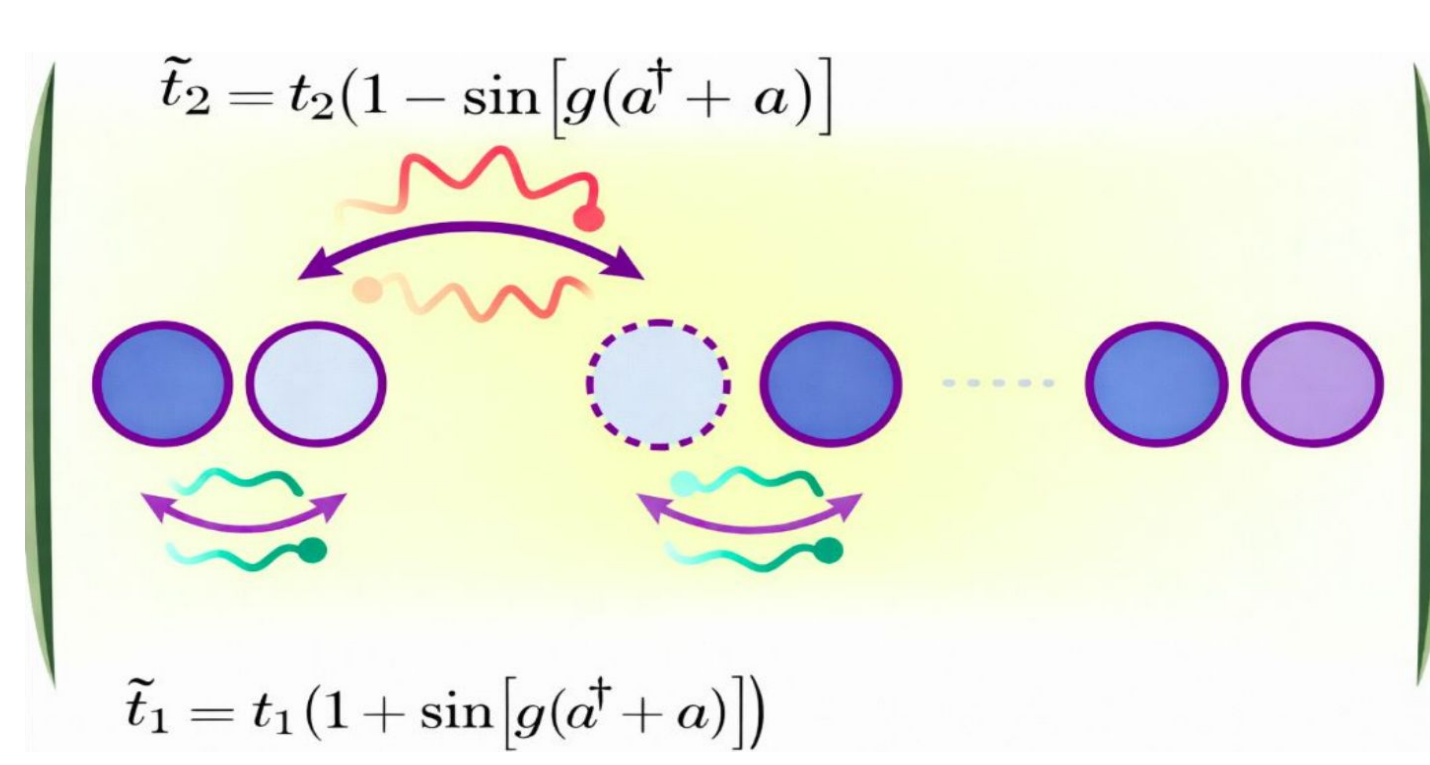
The light-matter coupling is introduced as a **complex phase** in the hoppings.



$$\begin{aligned} \tilde{t}_1 &= t_1 e^{i \frac{g}{\sqrt{L}} b_0 (a + a^\dagger)} \\ \tilde{t}_2 &= t_2 e^{i \frac{g}{\sqrt{L}} (1 - b_0) (a + a^\dagger)} \end{aligned}$$

Peierls Displacement:

Real intracell and intercell hopping amplitudes are sinusoidally modulated **out of phase**.



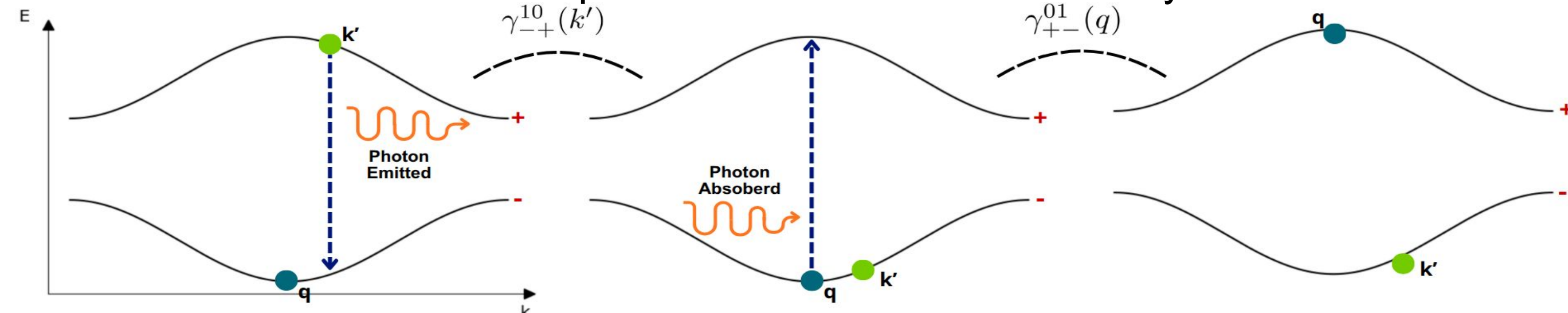
$$\begin{aligned} \tilde{t}_1 &= t_1 (1 + \sin [g(a^\dagger + a)]) \\ \tilde{t}_2 &= t_2 (1 - \sin [g(a^\dagger + a)]) \end{aligned}$$

Simplified Model

The spectral density $A(\mu, \omega)$ and Green's function are determined by the **single-photon transitions** of the coupled Hamiltonian:

$$H = \sum_{\alpha, \beta, n, m} \gamma_{\alpha\beta}^{nm} c_\alpha^\dagger c_\beta |n\rangle \langle m|$$

where $|n\rangle$ and $|m\rangle$ denote the photonic Fock states of the cavity.



The scattering process that dresses the electron consists of three stages:

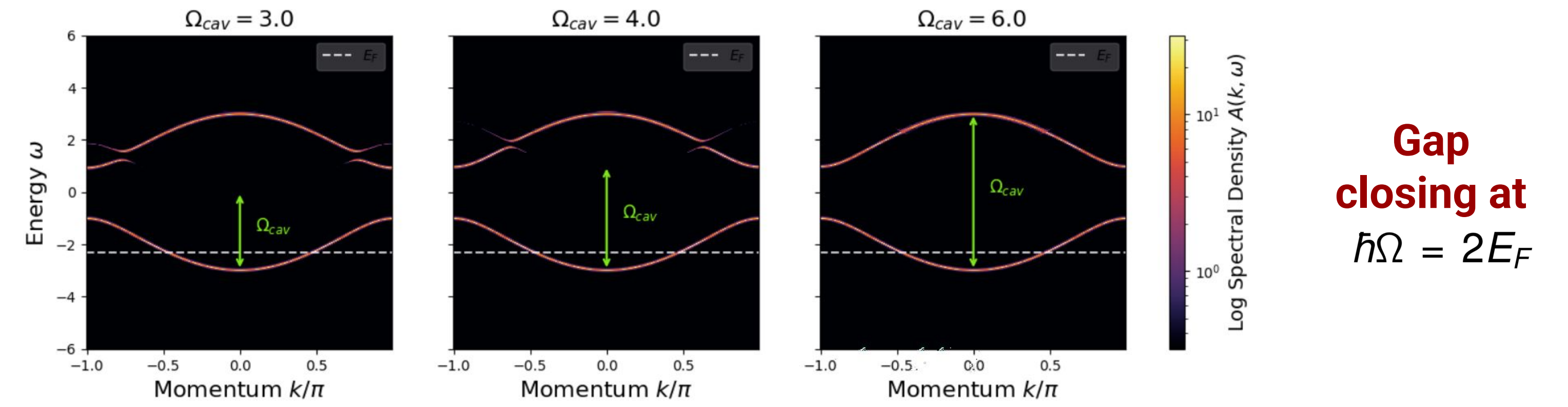
- **Injection (0 photons):** An electron is injected into an empty upper energy level $\mathbf{k}'$.
- **Emission (1 photon):** The electron decays to an unoccupied lower energy level (strictly outside the Pauli-blocked Fermi sea), emitting a cavity photon.
- **Absorption & Dressing (0 photons):** The photon is reabsorbed, exciting an electron $\mathbf{q}$ into an upper level.

References

- [1] Olesia Dmytruk and Marco Schiró. Controlling topological phases of matter with quantum light. *Communications Physics*, 5:271, 2022.
- [2] Beatriz Pérez-González, Gloria Platero, and Álvaro Gómez-León. Quantum origin of anomalous floquet phases in cavity-qed materials. *Communications Physics*, 7:419, 2024.

Results

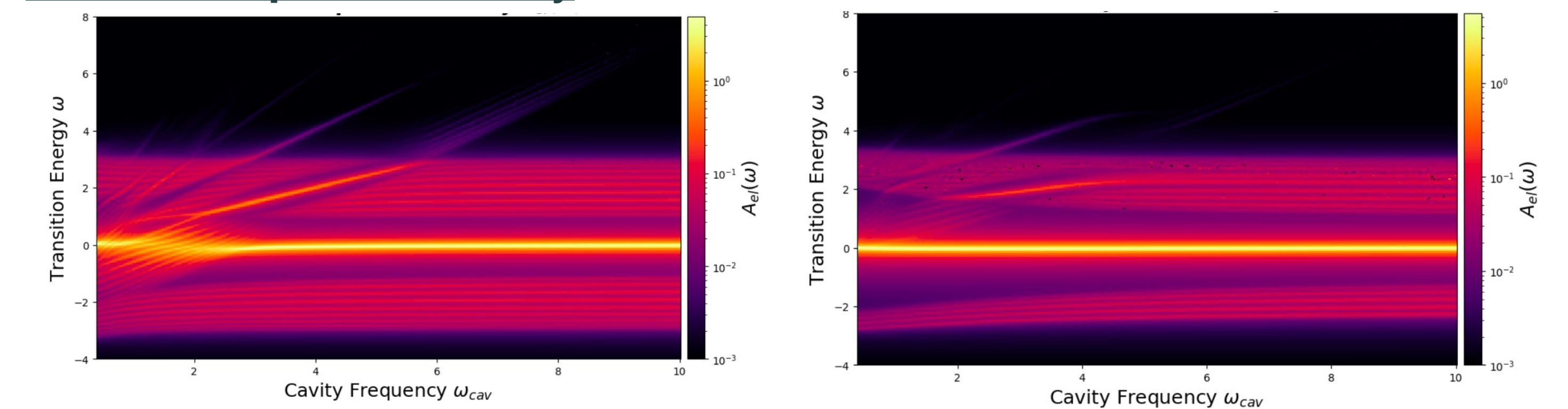
Spectral Density in k-Space



From the simulated spectral density $A(k, \omega)$ we observe three distinct regimes:

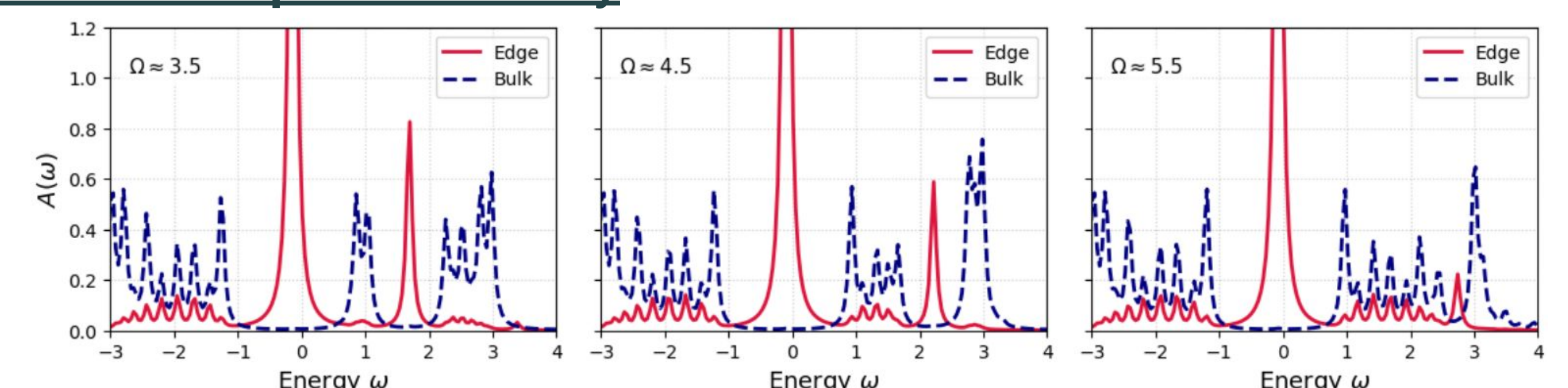
- **Resonant Regime:** Strong light-matter coupling induces an anticrossing between the upper band and the photon-dressed replica, opening a gap.
- **Intermediate Regime:** As Ω_{cav} increases, the replica shifts to higher energies, weakening the resonance (due to virtual particle-hole excitations) and starting to close the gap.
- **High-Frequency Regime :** Far from resonance, the replica is pushed out of the relevant spectrum. The gap vanishes, restoring the original band structure.

Electronic Spectral Density



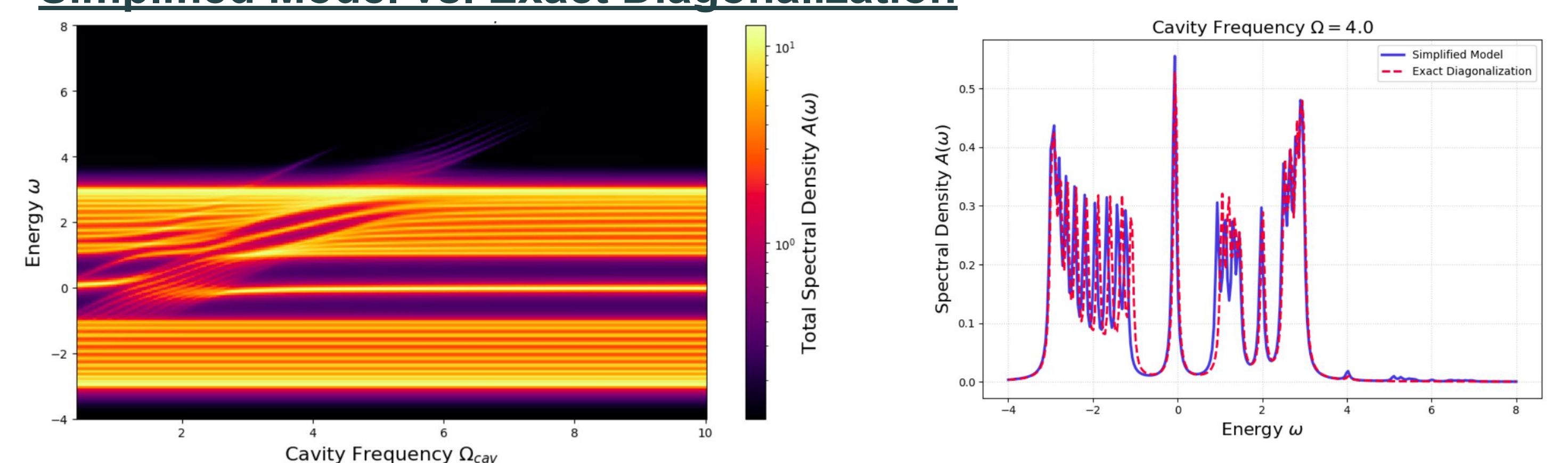
The band structure as a function of cavity frequency, computed via Exact Diagonalization, reveals two features: the intrinsic SSH gap and a photon-induced π -gap. While the standard SSH edge state persists at $\omega = 0$, a new **anomalous edge state** emerges crossing the π -gap, indicating a Floquet-like topological phase. However, many-body effects are critical: increasing the filling of electrons narrows the induced gap, causing the anomalous state to merge and vanish into the bulk.

Edge vs. Bulk Spectral Density



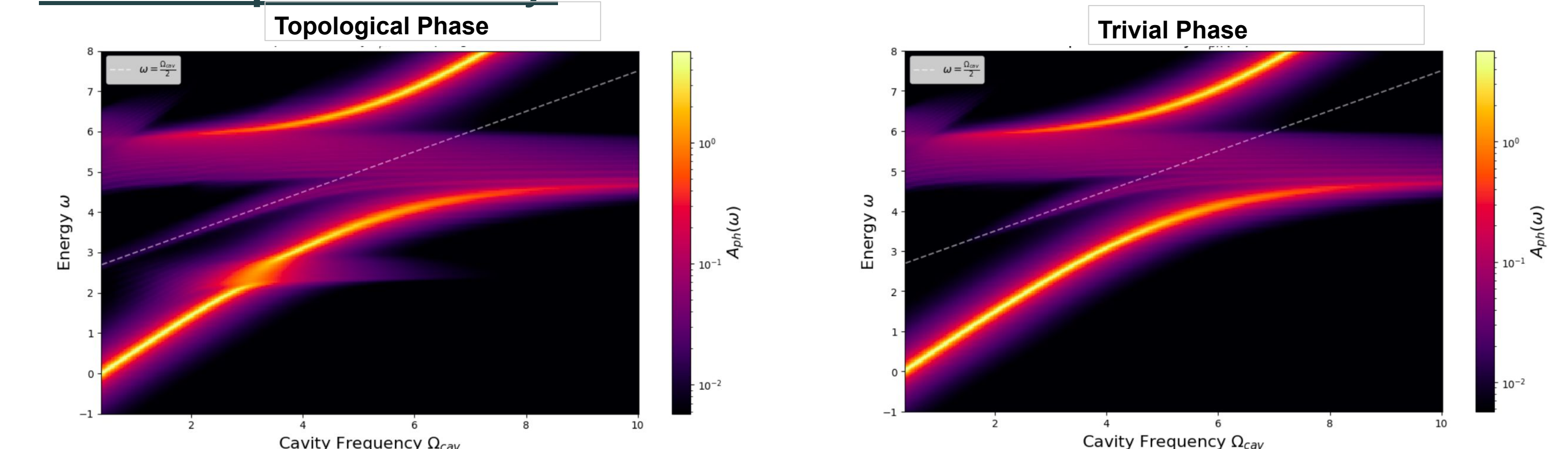
Spectral Signatures: Red peaks inside the blue bulk gaps identify the anomalous edge states. As cavity frequency increases, the boson-induced gap narrows and closes. The anomalous edge state loses its topological protection, hybridizes with the bulk bands, and becomes a delocalized state.

Simplified Model vs. Exact Diagonalization



The spectral density derived from the simplified model effectively captures the light-induced topological features. To validate our analytical approach, we compared its predictions with Exact Diagonalization (ED) results at a fixed cavity frequency. This strong agreement proves the simplified model is an accurate and efficient framework for exploring light-induced topological phenomena.

Photonic Spectral Density:



The bare cavity boson initially appears as a sharp pole. Coupling to the electronic system mixes this mode with a continuum of electron-hole excitations, leading to the formation of **hybrid polaritonic states** and a lifetime broadening of the boson. The trivial phase shows a single continuum of bulk transitions, whereas the topological phase features a dual continuum due to additional edge state transitions.

Conclusions

Coupling an SSH chain to a cavity mode creates a photon-induced π -gap and anomalous edge states. However, due to Pauli blocking, this gap closes and the edge states lose their topological protection, merging with the bulk as electron filling increases. Our simplified analytical model accurately captures these light-induced topological transitions, showing strong agreement with Exact Diagonalization.